

Andrew Boldi

Solving hard science problems with machine learning

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Education

UC Berkeley

Expected June 2027

BS in Chemical Biology & BA in Computer Science

- Fall '26:** Deep Neural Networks (CS 182), Computer Vision (CS 180), Physical Chemistry (Statistical Mechanics), Molecular Biology Lab
- Chemistry:** Computational Chemistry, Physical Chemistry (Quantum Mechanics), Honors Organic Chemistry I/II, Honors General Chemistry I/II
- Programming:** ML (CS 189), AI (CS 188), Data Structures & Algorithms (CS61B), Computer Architecture (CS 61C), Discrete Math & Probability Theory, Structure & Interpretation of Computer Programs, Designing Information Devices & Systems
- Sciences:** Macromolecular Biology, Chemical Biology (Biochemistry), Biological Inorganic Chemistry, Human Physiology, Mechanics, Electricity & Magnetism, Chemical Engineering Process Analysis, Mathematics & Statistics for Chemical Engineering

Work Experience

AI Intern

South San Francisco, CA

Soley Therapeutics: Molecular Machine Learning under direction of Dr. Ankur Gupta

May 2025 - Present

- Full-time summer 2025 and 2026 intern; received return offer and continued projects part-time during the academic year
- Filed patent, submitted manuscript**, and led development for a new explainability technique for molecular machine learning models. Demonstrated usefulness in internal lead optimization
- Leading development for computer vision explainability for cell images
- Model Finetuning: Finetuned hundreds of models with public and proprietary data to predict various molecular target candidate profiles (TCPs) to accelerate drug discovery
- Generative Chemistry: Developing generative AI chemistry models for novel generation and lead optimization of therapeutics
- Foundation Model Architecture Research: Implemented and benchmarked several ideas for better molecular property prediction foundation models; refactored codebase for existing foundation model
- Model Benchmarking: Implemented automated framework for configuration generation, model finetuning, and cross-fold validation for several foundation models across 22 datasets (TDC ADMET group)
- Automation: Automated CDD (Collaborative Drug Discovery) protocol creation via Selenium + Python, completing a task projected to take 100+ manual hours in 5-6 hours
- Product Integration: Integrated the above projects with UI/Infra team
- Tools used: PyTorch, PyTorch Lightning, Hydra/OmegaConf, CDD Vault, AWS EC2, S3, CodeBuild

Research Experience

Summer Research Intern

Pasadena, CA

Caltech: Professor Brian M. Stoltz Lab Group under Direction of Dr. Scott Virgil

June 2023 - August 2023

- Completed 16 consecutive reaction steps in the total synthesis towards an intermediate of a diterpenoid
- Developed software application for cataloging catalysts & ligands ([StremLigand](#) [🔗](#))
- Performed computational modeling (ORCA) for determining selectivity in a late-stage stereoinversion
- Used lab techniques including HPLC, LC/MS, GC/MS, LC-TOF, NMR, gCOSY 2D NMR, TLC, extraction, column chromatography, rotary evaporator, distillation, dry box

Summer Research Intern

Pasadena, CA

Caltech: Professor Brian M. Stoltz Lab Group under Direction of Dr. Scott Virgil

June 2022 - August 2022

- Completed the 10-step synthesis of a fragment in the total synthesis of korormicin
- Completed the 8-step synthesis of a fragment in the total synthesis of ineleganolide

- Conducted computational modeling of key intermediates to identify optimal synthesis routes using ORCA
- Designed and 3D printed broken HPLC autosampler part resulting in maintenance cost savings (~\$250/broken part) for use in the lab and other neurobiology labs at Caltech

Assistant Researcher

Remote

Caltech: Center for Catalysis and Chemical Synthesis under Direction of Dr. Scott Virgil

July 2020 - May 2022

- Developed software program for remote automation of robotic autosamplers (Python, VBA, M68k assembly language)
- Reverse-engineered M68K Assembly code to write more efficient functions for use in lab
- Used program to screen rare earth metal (lanthanide)-ligand catalysts for an asymmetric reduction

Research Summaries & Proposals

Synthetic Efforts Towards the Total Synthesis of Scabrolide B

August 2023

[2023 Summer Research](#) 

Synthesis of Korormicin and Ineleganolide Fragments

August 2022

[2022 Summer Research](#) 

Computational Total Synthesis Project Proposal

March 2024

[Deep Reinforcement Learning for Total Synthesis](#) 

Skills

Programming:	Python, Java, C/C++, Rust, Svelte, JS/CSS/HTML, M68k Assembly, Swift, LaTeX
ML & Tools:	PyTorch, PyTorch Lightning, TensorFlow, Hydra/OmegaConf, Selenium, AWS (EC2, S3, CodeBuild), Git/GitHub, Slurm, Linux
Chemistry:	Multistep organic synthesis, laboratory characterization (HPLC, LC/MS, NMR), computational modeling (ORCA, DFT), RDKit, AutoDock Vina
Languages:	English, Armenian, Spanish

Projects

Molecular Optical Rotation Prediction with Machine Learning

[Trennen](#) 

- Used T-Net model architecture to predict the rotation direction of optically active molecules; achieved 63% accuracy (50% random baseline)
- Distilled 8M+ molecules down to ~27,000 optically active molecules using PubChem's API, RDKit, and Openbabel
- Explored computing polarizability tensors with ORCA at the B3LYP/def2-SVP level of theory (DFT)
- Tools used: Tensorflow, Pubchem REST API, Openbabel, RDKit, ORCA

Scenic Navigation Algorithm with Deep Learning

[Pathfinder](#) 

- Developed navigation algorithm (modified A*) to find scenic routes in San Mateo County
- Developed web-scraping program with Selenium to automatically extract 250K+ Google Street View images from San Mateo County
- Used a pretrained semantic segmentation model for inference on 250K+ images with Google TPUs
- Accounted for curvature of the earth with mathematical expressions for better heuristic estimation
- Processed road connections and intersections with United States Census Bureau's TIGER/Line Shapefiles
- Tools used: Python, BDD100K Dataset, Google TPUs

Pacman with Bitcoin Miner (Top Project in Berkeley Course)

[Video Demo](#) 

- Built a Pacman game with a built-in Bitcoin miner with a project partner in 3000+ lines of Java code
- Selected as one of the top 15 projects in a class of ~1400 students for a data structures & algorithms course from UC Berkeley (CS61B, Spring 2024)

Zipf's Law Analysis in Large Corpus Texts

[Zipfai](#) 

- Analyzed Zipf's law (an empirical language processing law) and demonstrated it with large corpus texts
- Downloaded and processed 200+ GB of text: all of Wikipedia, manually web-scraped Project Gutenberg, etc.
- Tools used: Python, curl, wget, requests

Math Competition Training Website

[AMC Trainer](#)

- Developed interactive website and backend with my brother Ethan Boldi to help students prepare for math competitions (grades 8/10/12 and invitational AIME)
- Actively used by 20K+ users in 50+ countries
- Tools used: Javascript, HTML, CSS, Cloudflare Workers, Cloudflare KV Database, Node.js

Polyphenolic Content Analysis in Tea

[Video Demo](#)

- Analyzed the concentration of polyphenolic molecules in tea and kombucha for a project in Chem 4B (General Chemistry)
- Used Fast Blue BB (diazonium salt) to form spectrographically active substrate
- Designed and 3D printed custom cuvette vial plate for high-throughput analysis
- Wrote Python program to make 3D plots of wavelength absorbances
- Skills used: Spectrophotometry

Representation of the Fibonacci and Prime Numbers as Music

[Fibprimes](#)

- Built a program in C to automatically write MIDI files with the pattern of Fibonacci and prime numbers
- Recognized on Y Combinator's Hacker News and earned 26 stars on GitHub
- Tools used: C, MIDI file format

Contextual Translation Extension

[Fanyi](#)

- Built a Chrome extension in Javascript for contextually translating words or sentences by double-clicking on words
- Used OpenAI's API to fetch contextual information
- Installed by ~100 users
- Tools used: Javascript, OpenAI API

Bible Audio Player

[Simple Bible Audio](#)

- Built an app in Swift to listen to the Bible in Armenian or English with a single playback control
- Downloaded by ~100 people
- Tools used: Swift, Python

Additional Experience

Physician Shadowing at Kaiser Permanente

Summer 2026

- Shadowed Kaiser physicians and engineers in general surgery, Magnetoencephalography (MEG), hospital medicine, and palliative medicine.

Code Mind Berkeley

Nov 2023 - Jan 2024

- Interdisciplinary research for high school students with UC Berkeley professors
- Built website with Svelte + SvelteKit + Typescript + SCSS + GSAP
- Developed research project proposals at the intersection of machine learning, analytical chemistry, physics, biochemistry, marine science, and medicine
- Developed web-scraping pipeline for generating leads in Python + Selenium, later rewritten in Rust
- Secured \$250K in client commitments (75 clients); wound down the company to focus on full-time studies

Y Combinator AI Startup School

June 2025

- Selected as one of 2000 top CS undergrads, master's, and PhD candidates in AI to attend conference including speakers Elon Musk, Satya Nadella, Sam Altman, Andrej Karpathy, Andrew Ng, and more
- Further invited as one of ~20 students to have dinner with YC general partner Aaron Epstein

Private Tutor

2018 - present

- Teaching 7 students math and piano weekly
- Taught 25+ students in the past in subjects including linear algebra and chemistry

STEMPower: Workshop Facilitator & Program Director

Jul 2021 - Jun 2023

- Led hands-on workshops through Zoom to promote scientific interest for grades 3-6
- Subjects taught include math, chemistry, biology, and physics

YAPA Kids: Math Tutoring Teacher & Program Director

May 2020 - Aug 2021

- Led and taught interactive math classes for enhanced understanding of school topics
- Math Co-Program director for over 460 students from May 2021 to August 2021

Special Interests

Music

[YouTube Channel](#) 

- Multiple 1st place wins at piano competitions such as US Open Music, Armenian Allied Arts Association, Crestmont Conservatory Piano, and Satori Young Pianist (30+ recognitions)
- Performed at Carnegie Hall and Merkin Hall in New York City as a winner of the Armenian Youth Talent Competition
- Special recognition twice as Most Promising Young Talent at Armenian Allied Arts Association Competition
- Honorable mention at US International Music Competition
- Featured performer at Junior Bach Festival in Berkeley, California (and another 20+ public performances)
- Conductor of 200+ person pep band at Carlmont High School football games (Fall 2022)
- Pianist for Carlmont High School Jazz Ensemble (2019-2023)
- Studying classical piano since the age of 7
- At 12 years old, completed the final advanced level of the MTAC Certificate of Merit Program in Piano with branch honors and highest performance scores for all repertoire and perfect score on music theory exam
- With artistic flair, passion to perform, natural ability to learn, and love of music, constantly looking to serve surrounding community by performing at local Armenian community events and senior homes

Rubik's Cube Puzzles

- Solved 3x3 in 9.42 seconds & have solved Rubik's cube blindfolded
- Skilled at rapidly solving 2x2, 3x3, 4x4, 5x5, 6x6, 7x7, Megaminx (12-sided), and Square-1 cubes